SHS

AIRs - LM in

Business Mathematics

Week 6: Module 6

GOVERNMENT PROPERTY
NOT FOR SALE

Business Mathematics

Week 6: Module 6

Target

People often use the words markup and margin interchangeably. Many people do not know that there is a difference between the two terms, and unfortunately, the confusion between the two terms can negatively affect the bottom line of a business. The confusion between the meaning of the two terms stems from the fact that the same inputs are used to calculate both markup and margin, and the two of them provide information about the same transaction. However, margin and markup are totally different things. They show different information and are accounted differently.

In your previous lesson, you are done computing for the Mark-on, Mark-down and Mark-up given the price of a product.

This module will provide you with information and activities that will help you differentiate mark-up from margin.

After going through this learning material, you are expected to:

1. Differentiate mark-up from margins (**ABM_BM11B S-Ih-3**)

Module 6 Margin vs Markup

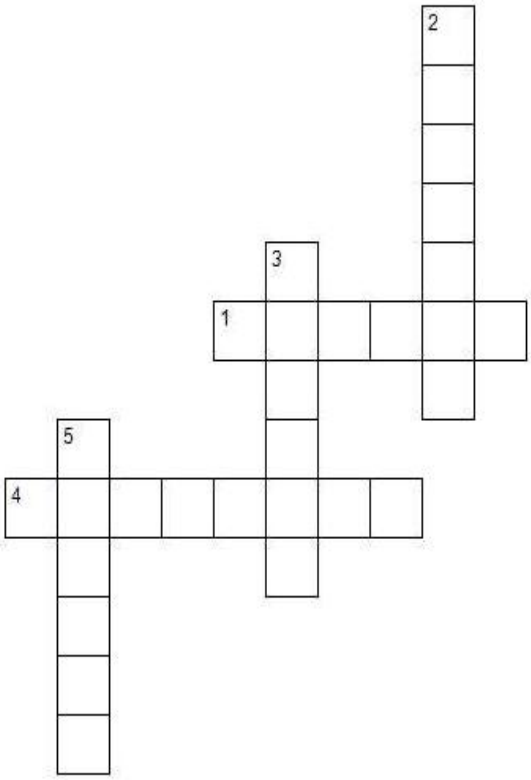
Jumpstart

For you to understand the lesson well, do the following activities. Have fun and good luck!

Activity 1: Fill it up.

Directions: Fill out the crossword puzzle below. The clue for each number is written on a box found at the right side of the puzzle. Each correct answer corresponds to two points (2) for a total of ten points (10). This contains some

keywords from your previous topic which will be used in this lesson. Write your answers legibly.



Across:

1. Amount added to the cost of goods

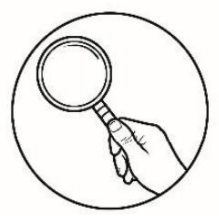
4. A reduction in price

Down:

2. Income generated from normal business operations.

3. Cost – Selling Price

5. Revenue – Expenses



Discover

Before going through the differences of markup and margin, it is important to first understand three terms which will come in handy when dealing with both margin and the markup.

These terms are:

- **Revenue**: This refers to the income earned after products or services are sold. Revenue reflects to all the money earned from the sales before any deductions have been made. Revenue is usually the top line in an income statement.
- **Cost of goods sold (COGS)**: This refers to all the expenses that the business incurs while making the products and delivering services. During the calculation of cost of goods sold, only variable costs are considered. Variable costs are those that are directly incurred in the production of goods and those that may vary depending on the amount of goods being produced. Examples of variable costs that are calculated as part of COGS include the cost of raw materials, manufacturing costs, product packaging, direct labor,

freight, and any other costs that can be directly attributed to making and selling the product. Fixed costs are not considered part of the cost of goods sold. Examples of fixed costs include rent, office expenses, and salaries of staff.

- **Gross profit**: This is the part of the revenue that remains after the expenses of manufacturing your products or delivering your products have been deducted. Gross profit is the difference between revenue and COGS.

What is Margin?

The margin, also referred to as gross margin, is a figure that shows the amount of revenue earned after the COGS has been deducted.

Margin can be expressed in Peso value or as a percentage. Margin is calculated by dividing the gross profit by the revenue.

Below is the formula for calculating margin:

$$\frac{(Revenue - Cost)}{Revenue} = Margin$$

To make the margin formula easier to understand, let's use an example to illustrate how it works.

Example:

Let's assume that an iPhone 11 is sold at PhP60,000 and costs the company Php30,000 to make.

The margin on the iPhone 11 would be:

$$\frac{(60{,}000 - 30{,}000)}{60{,}000} = 0.5\ Margin$$

Alternately, you can express the margin as a percentage by multiplying the figure above with 100.

$$0.5\ x\ 100 = 50\%\ Margin$$

In our example above, the margin for an iPhone11 is 50%.

This means that 50% of the total revenue is kept by the company, while the other 50% of the revenue covers the cost of producing iPhone 11.

From this, we can say that margin is a measure of how much of every peso earned in revenue is kept by the company after deducting expenses.

In our example, for every peso made in sales, the company retains Php0.50.

The higher the margin, the greater the portion of revenue the company keeps after making sales.

The gross margin is a very important metric when evaluating the financial performance of a company because it tells whether the company is making or losing money on sales, which is a very crucial aspect of business, since a business that is not making money on sales is failing. In addition, the gross margin is a useful indicator of how efficient the management of the company is in using supplies and labor in the production process.

For a company that has a very low gross margin, there are two major approaches for improving this key metric. The first one is by increasing the price of the products or services, while the second is by reducing the cost of production. None of these two approaches is easy. A price increase in a bid to increase profit margin can result in a reduction in sales. If the sales become too few, the business might be unable to bring enough revenue to cover operating costs. Therefore, before increasing the price, the business needs to consider factors such as supply and demand for the product, competition from the other businesses, inflation rates and so on.

The second option for companies that want to increase their gross margin is to reduce the variable costs associated with producing their product. For this to happen, the company needs to either reduce the cost of acquiring materials or make the production process more efficient. A great way of cutting costs on materials is to take advantage of volume discounts. By buying more material from a supplier at a go, you are more likely to receive discounts. Alternatively, you might opt to look for a less costly supplier.

You should be careful when doing this since low prices on materials might mean lower quality materials. If you decide to reduce your production cost by making your production process more efficient, you should also take care to ensure that the quality of goods is not compromised.

What is Markup?

Just like the margin, markup also analyzes the profit made after making a sale. However, markup looks at gross profit as a function of the cost of goods sold, rather than revenue. In other words, whereas you divide the gross profit by revenue to calculate margin, you have to divide the gross profit by the COGS to determine the markup. You can think of the markup as the extra percentage on top of the cost of production that you charge your costumers.

Markup can be calculated using the following formula:

$$\frac{(Revenue - COGS)}{COGS} = Markup$$

Once again, let's use the example from above where it takes Php30,000 to produce an iPhone11, which is then sold at a price of Php60,000. Here is how we would calculate the markup.

$$\frac{(60{,}000 - 30{,}000)}{30{,}000} = 1\ Margin$$

Alternatively, you can express the markup as a percentage by multiplying the figure above with 100.

$$1\ x\ 100 = 100\ Markup$$

The markup in this case is 100%, which means that the iPhone11 were sold for 100% more than what it cost to produce them. In other words, the selling price is double the cost of production.

Markup is a measure of how much more you sell a product compared to what it cost you to produce the product. Just like the margin, the higher the markup, the greater the portion of revenue the company keeps after making a sale.

Markup can also be described as the factor by which you multiply the cost of production to come up with a selling price. This can be expressed as:

$$Markup\ x\ Cost = Selling\ Price$$

The Difference Between Margin and Markup

As you might have realized by now, margin and markup are like two sides of the coin. They describe the same thing, but they provide different perspectives.
The margin shows the relationship between gross profit and revenue, while markup shows the relationship between profit and the cost of goods sold.

Aside from showing different perspectives, there are some other key differences between margin and markup, which include:

- Having a markup on your products ensures that your business is making a profit with each sale and provides a way of quantifying that profit.
- Markup is a great tool in the initial stages of a business since it helps you to better understand how cash flows into and out of your business. This can be very useful in helping you locate efficient points and bottlenecks within your business.
- Margin, on the other hand, is a precise and reliable tool for calculating profits and provides a clear picture of how sales are impacting your company's bottom line.

The difference between margin and markup is that the margin is sales minus the cost of the goods sold, while markup is the amount by which the cost of a product is increased in order to derive the selling price. A mistake in the use of these terms

can lead to price setting that is substantially too high or low, resulting in lost sales or lost profits, respectively. There can also be an inadvertent impact on market share, since excessively high or low prices may be well outside of the prices charged by competitors.

Margin vs Markup – Which should be used?

Generally, most small businesses, and especially retailers, depend on markup to set prices for their products. However, when it comes to recording financial information about a business, an accountant, bookkeeper or accounting software will be more interested in the margin rather than the markup.

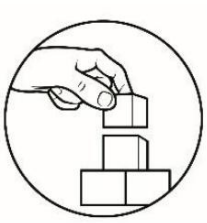
If you use markup in the place of margin, you will end up with bungled accounting numbers, which might make you think that your business is making more money than it is actually making. Therefore, while both can be used to determine how to price your products, you should stick to the gross margin when it comes to accounting, because it is a more accurate representation of the profit your business is making.

Wrapping up

Understanding the relationship between margin and markup as well as the difference between the two is very important for every business owner. Confusing between the two messes up your accounting and may even result in your business losing money without your knowledge.

On the other hand, knowing the difference between the two terms and how they are related with each other helps in setting the right goals for your business and implementing short and long term strategies for your business.

Understanding margin and markup also helps you to properly price your products. It allows you to competitively price your products while ensuring that you are not leaving any revenue on the table.



Explore

Here's an enrichment activity for you to work on to master and strengthen the basic concepts you have learned from this lesson.

CATEGORY	**Excellent 50 points**	**Very Good 40 points**	**Good 30 points**	**Fair 20 points**
Content	Covers topic in-depth with details and examples. Subject knowledge is excellent	Includes essential knowledge about the topic. Subject knowledge appears to be good.	Includes essential information about the topic but there are 1-2 factual errors.	Content is minimal or there are several factual errors.
Graphics	All graphics are related to the topic and make it easier to understand.	All graphics are related to the topic and most make it easier to understand.	All graphics relate to the topic.	Graphics do not relate to the topic.
Attractiveness	Makes excellent use of color, graphics, effects, etc. to enhance presentation	Makes good use of color, graphics, effects etc. to enhance presentation.	Makes use of color, graphics, effects, etc. but few detract from the presentation content.	Use of font, color, graphics, effects, etc. but most detract from the presentation content.
Mechanics	No misspellings or grammatical errors.	Minimal misspellings and/or mechanical errors.	Frequent misspellings and/or grammatical errors.	Major misspellings or grammatical errors.

Gauge

Directions: Read each item carefully. Use separate sheet for your answers. Write the letter of the best answer for each test item.

1. Which of the following best describes Markup?
   A. It determines the percentage of your profit.
   B. It is the amount used in producing a product.
   C. It helps in determining the selling price of a product.
   D. It shows the amount of revenue earned after the COGS has been deducted

2. Which of the following best describes a Margin?
   A. It determines the percentage of your profit.
   B. It is the amount used in producing a product.
   C. It helps in determining the selling price of a product.
   D. It shows the amount of revenue earned after the COGS has been deducted

3. This can be used in computing for the amount to be added in the cost production to obtain the selling price of a product.
   A. COGS
   B. Margin
   C. Markup
   D. Revenue

4. Which of the following best describes the difference of Margin and Markup?
   A. Margin computes for the profit from the product sold while Markup computes for the selling price of a product.
   B. Margin computes for the selling price of a product while Markup computes for the profit from the product sold.
   C. Margin shows the relationship between gross profit and cost of goods, while markup shows the relationship between profit and the revenue.
   D. Margin shows the relationship between gross profit and revenue, while markup shows the relationship between profit and the cost of goods sold.

5. What will be the percentage margin of a product sold at Php1,500 with a COGS of Php900?
   A. 4%
   B. 6%
   C. 40%
   D. 60%

6. If a product was sold at Php35,000 which was produced by using 30% of its selling price, how much was the markup?
   A. Php10,500
   B. Php15,000
   C. Php21,500
   D. Php25,100

7. This pertains to the amount of money spent in producing a product.
   A. COGS
   B. Margin
   C. Markup
   D. Revenue

8. This can also be described as the factor by which you multiply the cost of production to come up with a selling price.
   A. COGS
   B. Margin
   C. Markup
   D. Revenue

9. What is the relationship of Markup and margin?
   A. Markup should be relatively higher than margin to gain profit.
   B. Markup should be relatively lower than margin to gain profit.
   C. Markup should be relatively higher than margin to lessen the production cost.
   D. Markup should be relatively lower than margin to lessen the production cost.

10. What happens when your markup is a percentage smaller than the amount of your margin?
   A. You will have losses.
   B. You will gain more profit.
   C. Your selling price becomes higher.
   D. The production cost of your product decreases.

Great job! You are almost done with this module.

Business Mathematics

Week 6: Module 7

Target

Margins has always played a vital role in the progress of businesses in the market. It helps not only in knowing one's percentage of income but, it also helps in quantifying important data needed in the accounting world. Having a strong background knowledge about margins will give you an edge as a businessman in the spectrum of the business world.

In your previous lesson, you are done differentiating markup from margin.

This module will provide you with information and activities that will help you describe how margins is used in sales.

After going through this learning material, you are expected to:

1. Describe how gross margins is used in sales (**ABM_BM11B S-Ih-4**)

Gross Margins in Sales

ACTIVITY: FIGURE IT OUT!

Exercise 1.1

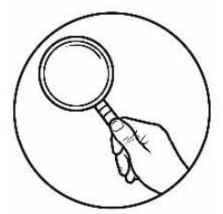
Lester owns a stone art business that designs and assembles stone art crafts for homes. He has two branches in La Union and has proven to be one of the most successful brands in his space. Here's what appears on Lester's income statement at the end of the year.

- Total sales: Php5,000,000
- Cost of Goods Sold: Php1,750,000
- Rent: Php500,000
- Utilities: Php50,000
- Other expenses: Php12,500

How much is his gross margin?

How about his gross profit margin?

____4. Margin	D. This is the part of the revenue that remains after the expenses of manufacturing your products or delivering your products have been deducted
____5. Markup	E. This is a figure that shows the amount of revenue earned after the COGS has been deducted



Discover

Gross margin is a company's net sales revenue minus its COGS. In other words, it is the sales revenue a company retains after incurring the direct costs associated with producing the goods it sells, and the services it provides. The higher the gross margin, the more capital a company retains on each peso of sales, which can then be used to pay other costs or to satisfy debt obligations. The net sales figure is simply gross revenue less the returns, allowances and discounts.

The formula for Gross Margin is:

Gross Margin = Net Sales – COGS
Where COGS = Cost of goods sold

The formula for Gross Profit Margin is:
Gross Profit Margin = (Net Sales – COGS)/Total Revenue
or = Gross Margin/Total Revenue

How to calculate Gross Margin

To illustrate an example of a gross margin calculation, imagine that a business collects 10 million pesos in sales revenue. Let us assume that the cost of goods consists of 1 million pesos spent on manufacturing supplies, plus 4 million pesos spent on labor costs. Therefore, after subtracting its COGS (10 million – (1 million + 4 million)), the company boasts a 5 million gross margin.

Examples of Gross Margin Calculations used in sales.

1. If a retailer had net sales of Php200,000 and its cost of goods sold was Php120,000, the retailer had a gross margin of Php80,000 or 40% of net sales (80,000/200,000).

2. A company earned Php550,000 in revenue by selling shoes, and the shoes created Php220,000 of labor and materials costs to produce. To calculate gross profit, the company subtracts cost of goods sold from revenue:

   Php550,000 – Php220,000 = Php330,000.

   To get the gross profit margin, divide gross profit by revenue:

   Php330,000/Php550,000 = 0.6 or 60%.

   This means that the direct costs of producing the product that the company sells consume 40% of its revenue, and that it has 60% of its revenue to cover indirect costs and create profit for the owners.

3. An online seller spent Php20,000 buying materials and products for his online business. He was able to sell his products for Php100,000. Unfortunately, Php10,000 of the sales were returned by customers and refunded. He would calculate his gross margin ratio like this:

$$\frac{90{,}000 - 20{,}000}{100{,}000 - 10{,}000}$$

   90,000 is the result of subtracting Php10,000 (value of returned sales) from Php100,000 (total sales).

   Upon further computation, the online seller still got .77777 or .78. This means that after paying his COGS, he still has 78 percent of his sales revenue to cover his operating costs.

4. For the month of July, 2020, A company earned revenue of Php37,210,000 by selling face shield and face masks costing Php25,194,500. What is the gross profit margin of the company?

   Solution:

$$\frac{37{,}210{,}000 - 25{,}194{,}500}{37{,}210{,}000} \approx 0.32 \; or \; 32\%$$

5. Calculate the Gross Profit Margin of a company whose goods sold and gross profit for the period are Php437,700,000 and Php121,150,000 respectively.

Solution:

Since the revenue figure is not provided, we need to calculate it first:
Revenue = Gross Profit + COGS
Revenue = Php437,700,000 + Php121,150,000
Revenue = Php558,850,000

Gross Profit Margin =

$$\frac{121{,}150{,}000}{558{,}850{,}000} \approx 0.22 \; or \; 22\%$$

What does the Gross Margin Tell you?

The gross margin represents the portion of each peso of the revenue that the company retains as gross profit. For example, if a company's recent quarterly gross margin is 35%, that means it retains .35 pesos from each dollar of revenue generated. Because COGS have already been taken into account, those remaining funds may consequently be channeled toward paying debts, general and administrative expenses, interest fees, and dividend distributors to shareholders.
Companies use gross margin to measure how their production cost relate to their revenues. For example, if a company's gross margin is falling, it may strive to slash labor costs or source cheaper suppliers of materials. Alternatively, it may decide to increase prices, as a revenue increasing measure. Gross profit margins can also be used to measure company efficiency or to compare two companies of different market capitalizations.

Why Does Gross Margin Matter?

Gross margin is generally important because it is the starting point toward achieving a healthy net profit. When you have a high gross profit margin, you are in better position to have a strong operating profit margin and strong net income. For a newer business, the higher your gross profit margin, the faster you reach the break-even point and begin earning profits from basic business activities.

This does not always mean a high margin is possible however. The pricing strategy and competition will ultimately drive how the margin reacts to the consumer buying habits. Nobody wants to capture the highest possible margin without sacrificing sales to maximize revenue.

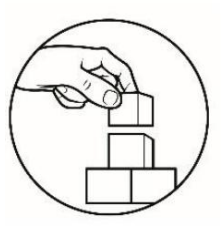
Margin-Based Pricing Strategies

The gross margins are often determined by pricing strategies. Typically, the way a product is priced is based on competitive market pricing. In other words, you will

price similar to the competition and you'll accept the standard margins while also attempting to market your product so that you drive sales.

In some instances, it pays to lower price than the market while also accepting a lower gross margin. The decreased margin can lead to increased sales as you offer the best price point. It can also backfire as competitors reduce prices, and everyone experiences a lower margin against similar sales trends.

Another strategy being used in the market is that of pricing higher than the market to maximize the margins. A high pricing strategy is often accompanied by a major branding campaign. In this case, the company is really selling the brand as much as the product so that it can achieve sales at the higher pricing. This strategy can work in some markets, but it does come with the initial risk of selling to a market that's comfortable buying at a lower price.



Explore

Here's an enrichment activity for you to work on to master and strengthen the basic concepts you have learned from this lesson.

Enrichment Activity 1: Remember Me.

Directions: Write three events in your life where you were able to use or experience the application of margin or gross margin in relation to sales. Write the specific event where it happened and describe how it was used. Use the table provided below.

	appropriate to audience.			
Convention	Evident control of grammar, mechanics, spelling, usage and sentence formation.	Sufficient control of grammar, mechanics, spelling, usage and sentence formation.	Limited control of grammar, mechanics, spelling, usage and sentence formation.	Minimal control of grammar, mechanics, spelling, usage and sentence formation.

Gauge

Directions: Read each item carefully. Use separate sheet for your answers. Write the letter of the best answer for each test item.

1. Which of the following describes the process of obtaining the gross margin?
   A. Net sales revenue minus cost of goods sold.
   B. Net sales revenue minus the net sales profit.
   C. Net sales revenue plus the cost of goods sold.
   D. The cost of goods sold minus the net sales revenue.

2. Which of the following is true about margins?
   A. The higher the gross margin, the more capital a company retains on each peso of sales.
   B. The higher the gross margin, the more debt a company retains on each peso of sales.
   C. The higher the gross margin, the lesser capital a company retains on each peso of sales.
   D. The higher the gross margin, the lesser debt a company retains on each peso of sales.

3. How much is the gross margin of an online succulent seller who has a COGS of Php7,500 and a revenue of Php10,000?
   A. Php1,500
   B. Php2,500
   C. Php7,500
   D. Php10,000

4. What is the gross margin percentage of a business owner who has spent a total of Php110,000 in producing all his products and has a total revenue of Php275,000?
   A. 40%
   B. 60%
   C. 140%
   D. 160%

5. If a business owner has a total COGS of Php23,000 and a total sale of Php84,000, what is the margin percentage of his business?
   A. 27.38%
   B. 38.27%
   C. 62.72%
   D. 72.62%

6. An online seller had total sales of Php54,000. Unfortunately, Php7,000 of the sales were returned by customers and refunded. If she had spent Php14,000 to produce all his product, what is her gross margin percentage?
   A. 25.21%
   B. 50.21%
   C. 70.21%
   D. 80.21%

7. A cellphone seller spent Php425,000 in purchasing his products. His total sales are Php900,000 but, Php50,000 worth of his products were defective and the customers demanded for a refund. How much is his total margin?
   A. Php375,000
   B. Php425,000
   C. Php850,000
   D. Php950,000

8. What does gross margin represent in a business?
   A. The portion of each peso of the revenue that the company retains as gross profit Margin
   B. The portion of each peso of the revenue that the company retains as net profit Margin.
   C. The portion of each peso of the revenue that the company retains as revenue.
   D. The portion of each peso of the revenue that the company retains as COGS.

9. What does a high gross margin profit mean in a business?
   A. The business has a weak operating profit margin and a weak net income.
   B. The business has a strong operating profit margin and strong net income
   C. The business has a weak operating profit margin but strong net income.
   D. The business has a strong operating profit margin but a weak net income.

10. What would be the possible result of decreasing your profit margin?
   A. Higher sales since you have better price sales
   B. Lower sales since your COGS will also decrease.
   C. Higher sales since your COGS will also decrease.
   D. Lower sales since your selling price will be lower.

Great job! You are almost done with this module.

Business Mathematics

Week 6: Module 8

Target

In the business industry, the supplier usually recommends the suggested retail price of an item. However, because there is a line of other people who are involved in bringing an item to the hands of the consumer, these middlemen expect their own share of profit also. Hence, the supplier normally gives them a special discount to grant them a small room for profit. This discount is referred to as trade discount.

In your previous lesson, you were taught on how gross margins is used in sales. It can be used to measure a company's efficiency. This also helps a company in assessing the profitability of its manufacturing activities.

This module will help you in learning the computation of single trade discounts and discount series.

After going through this learning material, you are expected to:

1. Compute single trade discounts and discount series.

Single Trade Discount and Discount Series

Jumpstart

For you to understand the lesson well, do the following activities. Have fun and good luck!

Activity 1: Read and Understand me.

Directions: Read the given passage below carefully and try to understand its meaning.

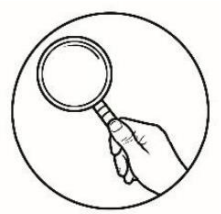
Almost every student nowadays owns a mobile device (ex: cellphone, tablet, iPad). But have you ever wondered about the process these goods go through before we finally buy them and lay ownership on them? Others may have bought it from the department store, a cellular phone company or from a stall inside the mall. These sources are called retailers. How these retailers got hold of the selection of cellular phones they had to choose from? If you have an iPhone, for instance, the iPhone store where you bought it received a variety of units from an official distributor in the Philippines. This official distributor will have links to the original source or supplier of the iPhone which in this case is Apple.

Activity 2: Rearrange Me

Directions: Arrange the following boxes below to complete the product cycle in the market. Use arrows to indicate chronological order. Use a separate sheet to draw your answers.

1.

2.



Discover

A **trade discount** is the amount by which a manufacturer reduces the retail price of a product when it sells to a reseller, rather than to the end customer. The reseller then charges the full retail price to its customers in order to earn a profit on the difference between the amount by which the manufacturer sold the product and the price at which it sells the product to the final customer.

There are different kinds of trade discount. Let us first talk about **single trade discounts.**

1. When you go to the market and buy meat from a regular vendor or "suki" (as we say in Filipino), you are sometimes given an **incentive** or **loyalty discount**.

2. When you buy an item which a vendor is willing to give out on installment, you may be able to convince him/her to give you a **cash discount** if you are willing to pay the full amount in cash.

3. During Christmas season and you shop for gifts for your friends and family, we convince sellers to give us a discount if we buy a stated minimum number of items. This is called a **volume discount.**

To compute for the **TRADE DISCOUNT**:

TRADE DISCOUNT = TRADE DISCOUNT RATE X SELLING PRICE

Examples:

1. My regular beef supplier gives me a 5% discount on my meat purchases every week. If the price of beef brisket this week is PhP375 per kilo and I am planning to buy 1.5 kilos, how much trade discount will I receive from this purchase?

   Solution:

   Total Selling Price = 1.5 x 375

   = PhP562.50

   Trade Discount = 0.05 x 562.50

   = PhP28.13

2. My mother admired a pair of earrings that her friend was offering for PhP24,000, payable in 6 months. She bargained to be given a discount and her friend agreed to give her 12% if she pays in cash. How much will my mother have to pay for the pair of earrings?

   Solution: Trade Discount = 0.12 x 24,000

   = PhP2,880

   Discounted Price = 24,000 – 2,880

   = PhP21,120

Equivalent Solution:

Since trade discount is 12%, customer pays 88% of the selling price, that is

Discounted Price = 0.88 x 24,000 = PhP21,120

My mother should pay her friend PhP21,120 in cash to avail the 12% discount.

3. For Christmas, Paul gave each of his 34 classmates a shirt with a unique design for their class remembrance. He found a supplier who gives 8% discount for a minimum of 25 shirts purchased. The total cost of the shirts Paul purchased amounted to PhP3,910. How much was the unit selling price of the shirts?

Solution:

100% – 8% = 92%

(The total price for the shirts Paul bought was only 92% of their original selling price.)

Original Selling Price = $\frac{3910}{0.92}$

= PhP4,250

Unit Selling Price = $\frac{4250}{34}$

= PhP125

The unit selling price of the shirts is PhP125.

Equivalent solution:

Discounted Price per shirt = $\frac{3910}{34}$ = 115

Paul paid PhP115 per shirt.

Unit Selling Price = $\frac{115}{0.95}$ = 125

The unit selling price is PhP125.

There are times when a supplier also designs discounts to encourage specific behavior from its clients. For instance, a supplier may want to give its distributor a discount of 20/10/5. This notation represents a trade discount series. To interpret, the supplier may, in addition to the 20% discount it gives the distributors, slash another 10% from the resulting discounted price if the

distributor purchases a set minimum volume of items and another additional 5% if the purchase is paid in cash. Is this equivalent to a 25% single trade discount?

Example:

If a popular shoe brand promotes this discount of 20/10/5 for a minimum volume of 250 pairs of shoes. Assuming that the unit selling price of the shoes is PhP2,500, compute the trade discount per pair of shoes that a distributor avails of under this scheme.

First Trade Discount = 0.20 x 2,500 = PhP500

Initial Discounted Price = 2,500 – 500 = PhP2,000

Or

Initial Discounted Price = (1 – 0.80) x 2,500 = PhP2,000

Second Trade Discount = 0.10 x 2,000 = PhP200

Second Discounted Price = 2,000 – 200 = PhP1,800

Or

Second Discounted Price = (1 – 0.10) x 2,000 = PhP1,800

Last Trade Discount = 0.05 x 1,800 = PhP90

Total Trade Discount = 500 + 200 + 90

= PhP790

The distributor will be able to avail of a total of PhP790 of trade discounts if he purchases a minimum of 250 pairs of shoes.

Equivalently, if distributor purchases a minimum of 250 pairs of shoes, he will pay an amount of:

Final discounted price = (1 – 0.05) x (1 – 0.10) x (1 – 0.20) x2,500

= 0.95 x 0.90 x 0.8 x 2,500

= 0.684 x 2,500

= PhP1,710 for each pair of shoes.

Total discount = 2,500 – 1710

= PhP790

Is this the same as a 35% single trade discount?

Note that the total trade discount availed of by the distributor did not amount to this much, but lower. A trade discount series of 20/10/5 is not equivalent to a 35% single trade discount.

To compute the equivalent single trade discount of 20/10/5,

Single Trade Discount Rate Equivalent = $\frac{790}{2500}$ = 0.316 = 31.6%

The trade discount series 20/10/5 is only equivalent to a single trade discount of 31.6%, not 35%.

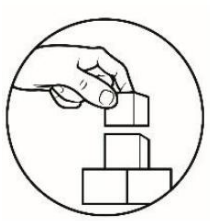
Alternate solution:

Single Trade Discount Rate Equivalent = 1 - (1-0.05) x (1-0.10) x (1-0.20)

= 1 - 0.95 x 0.9 x 0.8

= 1 - 0.684

= 0.316 = 31.6%



Explore

Here's an enrichment activity for you to work on to master and strengthen the basic concepts you have learned from this lesson.

Enrichment Activity 1: Solve me!

Directions: Below are real life scenarios in which single trade discount and discount series were used. Solve this using a separate sheet of paper. Use calculator if needed.

1. A manufacturer sells fishing boats to dealers for Php94,500 with discounts of 40/10/5.

   a. Determine the net price of a kayak.

   b. How much is the total trade discount?

   c. What is the equivalent single trade discount for this transaction?

2. What is the equivalent single trade discount for a trade discount series of 30%, 10.5%, 12 $\frac{1}{3}$%, and 3%?

3. A retail kiosk in San Fernando has listed a pair of sunglasses for Php4,760 less 30%. A department store in the shopping mall lists the same model for Php4,410 less 20%, less an additional 15%. What additional rate of discount must the kiosk give to meet the department store price?

Deepen

Directions: Complete the given table of values below. Use the provided values in solving for the missing numbers. Show your complete solutions.

What you need:

Short bond paper, calculator and pen.

Original Selling Price	Discount Rate	Discount	Sale Price	Sale Price Rate
PhP158.75	5%			
PhP865.00	12.5%			
PhP1,300.00		PhP104.00		
PhP3,650.00		PhP803.00		
PhP725.00			Php522.00	
PhP630.00			PhP582.75	
	3%		Php945.75	

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BUSINESS MATH QUARTER 3

Most Essential Learning Competencies	Output To Be Submitted	Date of Completion
Quarter III Week 1 ***2 Fractions/ decimals and percentage*** **February 21-25, 2022** ✓ Give real-life situations to illustrate fractions, decimals, and percent (ABM_BM11FO-Ic-4) ✓ Solve problems involving fractions, decimals and percent (ABM_BM11FO-Id-5)	**Worksheet for Week 1 -2 (Reference: Activity Sheet Quarter III)** Instructions: Answer the following activities. Module 1-2 . Gauge. Activity 8. Convert Me! (A, B, C, D, E, F) o Write all your answers on yellow paper. o Write the activity title o Write the quarter number and week number on the top right side of the paper.	**March 3, 2022**
Quarter III Week 3 – ***Key Concepts of Ratio and Proportion*** **February 28 – March 4, 2022** ✓ Identify the different kinds of proportions and write examples of real-life situations for each	**Worksheet for Week 3 (Reference: Activity Sheet Quarter III)** Instructions: Answer the following activities. Module 3. Explore. Activity 1. #s 1-5	**March 7, 2022**
Quarter III Week 4 ***– Key Concepts of Ratio and Proportion*** **March 7-11, 2022** ✓ Solve problems involving direct, inverse and partitive proportion (ABM_BM11RP-If-4)	**Worksheet for Week 4 (Reference: Activity Sheet Quarter III)** Instructions: Answer the following activities. Module 4. Gauge #s 1-15	**March 14, 2022**
Quarter III Week 5 ***Key Concepts of Buying and Selling*** **March 14-18, 2022** ✓ Differentiate Mark-on, Mark-down, and Mark-up obtain Mark-on, Mark-down and Mark-up of a product (ABM_BM11BS-Ig-1)	**Worksheet for Week 5 (Reference: Activity Sheet Quarter III)** Instructions: Answer the following activities. Module 5a. Explore. Activity 1. Can You Tell? #s 1-10 Module 5b, Explore. Activity 1. Complete Me! Letters – A, B, C	**March 21, 2022**
Quarter III Week 6 ***Key Concepts of Buying and Selling*** **March 21-25, 2022** ✓ Differentiate Mark-up from margins (ABM_BM11BS-Ih-3) ✓ Describe how gross margins is used in sales (ABM_BM11BS-Ih-4) ✓ Compute single trade discounts and discount series	**Worksheet for Week 6 (Reference: Activity Sheet Quarter III)** Instructions: Answer the following activities Module 6. Gauge #s 1-10 Module 7. Gauge #s 1-10 Module 8. Enrichment. Activity 1. Solve Me #s 1-3	**March 28, 2022**

Quarter III Week *7 Key Concepts of Buying and Selling* **March 28- April 1, 2022** ✓ Differentiate profit from loss (ABM_BM11BS-Ii-6) ✓ Illustrate how profit is obtained and how to avoid loss in a given transaction (ABM_BM11BS-Ii-7)	**Worksheet for Week 7 (Reference Activity Sheet Quarter III)** Instructions: Answer the following activities Module 9. Explore #s 1-5 Module 10. Gauge #S 1-10	**April 4, 2022**
Quarter III Week *8 Key Concepts of Buying and Selling* **April 4-8, 2022** ✓ Determine the break-even point ✓ Solve problems involving buying and selling products (ABM_BM11BS-Ij-9)	**Worksheet for Week 8 (Reference: Activity Sheet Quarter 1II)** Instructions: Answer the following activities Module 11. Explore. Activity 1. Is it a Fact or a Bluff?	**April 11, 2022**
Quarter III Week 9 Key Concepts of Buying and Selling **April 11-13, 2022** ✓ Compute interest specifically as applied to mortgage, amortization and on services/ utilities and on deposits and loans	**Worksheet for Week 8 (Reference: Activity Sheet Quarter 1II)** Instructions: Answer the following activities Module 13. Gauge # 1-15	**April 18, 2022**